Article

Verified Localized Episodic Grounding for Auditable Network Traffic Analysis

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Abstract: Memory-augmented large language models (LLMs) can assist few-shot network intrusion triage by summarizing historical flows into natural-language rules without parameter updates, yet local rules risk label hallucination and cross-flow over-generalization. We propose Verified Localized Episodic Grounding (VLEG), an auditable consolidation and hybrid-inference framework that verifies candidate reflections against their labeled source records within K attempts and screens admitted rules with a deterministic counterexample compiler. In a 16-scene symbolic sanity check, bounded verification ($K = 3$) eliminates source-label inversions (100% observed agreement), performing comparably to raw retrieval at 70B (91.67% versus 89.58%) while trailing at 7B (75.0% versus 87.5%). On a five-seed, 32-flow benchmark over 1000 deduplicated UNSW-NB15 test fingerprints, unverified single-step reflection mislabels 95.0% (76/80) of malicious support flows as benign, causing test attack recall to drop to 6.17% (42.66% Macro-F1). Multi-step verification in VLEG restores 100% source-label agreement (158/158) and raises Macro-F1 to 59.68%, but trails raw retrieval (63.92% Macro-F1) because 59.2% (93/157) of compiled rules match contrary-label flows within their 32-flow seed (72.0% across all 160 pooled flows), inflating the false positive rate (FPR) to 48.78%. Gating out rules with within-seed counterexamples (VLEG-CG) reduces the FPR to 37.86%–39.43%, recovers overall Macro-F1 to 63.32%–63.55%, and outperforms raw retrieval (64.60% versus 63.03% Macro-F1) on the un-gated subset where zero-counterexample rules are retained.

Keywords: episodic memory; hallucination mitigation; large language models; network traffic analysis; intrusion detection; rule over-generalization; verified memory consolidation

1. Introduction

1.1. Background and Motivation

Large language models (LLMs) can use examples in the prompt to support classification and incident analysis without parameter updates [1, 2]. For network traffic analysis and Security Operations Center (SOC) intrusion triage [3–6], historical flow records and incident summaries offer critical evidence for interpreting new observations. Whereas traditional supervised classifiers such as Random Forest (RF), Gradient Boosting Decision Trees (GBDT), and CART decision trees achieve high binary throughput when trained on tens of thousands of labeled flows, analysts investigating novel or scarce alerts frequently seek explicit, human-readable verbal rationales linking a target flow to curated historical precedents. Their usefulness depends on retaining relevant conditions and exceptions while avoiding unsupported conclusions. Finite and imperfectly used context windows make the selection and organization of this historical evidence important [7, 8].

Persistent-memory systems provide access to experience beyond the current prompt [9–13]. Raw records preserve instance-level detail, while reflections and summaries supply compact guidance. However, a natural-language summary can hallucinate a class label, omit a critical exception, or turn an incidental attribute into a general rule. This paper examines that trade-off using controlled symbolic sanity checks and a public network-flow benchmark.

The memory-design problem is particularly visible when a model must relate a new observation to a small collection of labeled examples. Keeping every detail can preserve useful exceptions, but the model must still infer which differences matter at test time. Compressing the same examples into natural-language rules shifts this

burden because inference can follow the stored guidance directly, even though the abstraction may be less faithful than the underlying records. The design decision therefore concerns both what is retained during consolidation and what evidence accompanies an abstraction when it is reused.

Crucially, source agreement and general validity are distinct requirements. A proposed explanation can correctly describe one record while asserting a condition that fails elsewhere. For example, an observed attribute or threshold may coincide with a malicious label in one flow without being sufficient to determine it across the broader traffic distribution. Checking the explanation against that single originating record cannot resolve the distinction. A useful memory system must consequently be evaluated beyond successful storage: the resulting memory should support predictions on reserved queries, and its explanatory claims should be examined against contrary-label counterexamples. This motivates measuring memory admission coverage, source-label consistency, within-seed and cross-seed counterexample rates, and downstream classification outcomes together rather than treating an admission rate as a complete measure of memory quality.

1.2. Memory Research and Study Positioning

Memory-augmented language models retain and reuse experience through episodic retrieval, local reflection, and semantic consolidation [14,15]. These approaches differ in their abstraction unit, correction signal, and inference evidence. Feedback-based agents already revise behavior using past trial experience [16,17]; global hypothesis methods also generate and verify abstractions across examples [18,19]. Our focus is the narrower combination of per-episode admission checking, evidence–reflection pairing, and deterministic counterexample screening. We compare its outcomes against raw retrieval, abstract memory, shallow symbolic decision trees, and supervised tabular ensembles rather than assuming that local verification or hybrid context is universally superior.

The choice between these approaches depends on what must be preserved and how stored knowledge will be reused. Episodic retrieval retains concrete examples; reflection converts local experience into reusable verbal guidance; semantic consolidation summarizes patterns across records. Hybrid inference combines an abstraction with retrieved evidence. Each can be useful under suitable conditions, and each introduces different failure modes when evidence is incomplete or an abstraction is applied outside its support.

The first research question of this study is whether checking a local reflection against its labeled source before admission prevents the label hallucinations of unverified reflection, and whether this gain requires multi-step verification rather than single-pass label disclosure. A second question is whether pairing that reflection with its source record is preferable to supplying either component alone. A third question is why 100% source-consistent reflections still inflate false alarms on real-world network flows, and how deterministic counterexample gating within the support bank affects both gated fallback queries and un-gated rule-retaining queries. Answering these questions requires controlled comparisons where retrieval selection is separated from displayed context and single-instance verification is separated from cross-episode specificity.

1.3. Proposed VLEG Framework and Counterexample Gate

We propose Verified Localized Episodic Grounding (VLEG) to check local abstractions before they enter memory while retaining the evidence from which they were derived. A generator proposes a local reflection from label-free historical attributes, and a verifier checks it against the labeled source episode. Rejected candidates receive diagnostic critique and undergo iterative revision within a maximum budget K, after which the first accepted candidate is stored alongside its source record. At inference, retrieved tuples provide both raw evidence and accepted reflections.

Generation and verification are separate calls to the same underlying model, and target test labels are strictly withheld from inference. While the admission check exposes direct contradictions or label inversions with respect to a source episode, acceptance remains a local model judgment rather than a guarantee of causal validity or cross-episode transfer. These limits define the questions examined experimentally.

The consolidation procedure operates on labeled historical episodes rather than on the target query. A proposed reflection summarizes a local condition or detection heuristic. If the verifier rejects it, the critique becomes available to the next generation attempt. The iteration budget K bounds the number of revision attempts, omitting any episode whose proposals remain rejected from the admitted memory bank. This creates an explicit trade-off between construction token work and the amount of historical evidence retained for inference.

Each admitted tuple keeps the episode and its accepted reflection together, allowing the inference prompt

to expose both. In our public-flow evaluation, we standardize feature-based example selection across all local conditions so that differences in downstream classification strictly isolate memory-consolidation and gating quality from retrieval variance.

Furthermore, to diagnose and mitigate the false-positive inflation caused by overly broad local heuristics, we pair VLEG with a deterministic counterexample compiler (Ψ) and a within-bank counterexample gate (VLEG-CG). During consolidation, Ψ compiles each admitted natural-language `IF ... THEN ...` reflection into an executable Boolean predicate over flow features and evaluates it against the remaining historical episodes in the same support bank. When a retrieved rule matches contrary-label historical flows, VLEG-CG strips the over-generalized verbal heuristic at inference and falls back to the verified raw flow evidence.

1.4. Experimental Evaluation and Summary of Contributions

We evaluate VLEG on a controlled 16-scene symbolic sanity check at 7B and 70B model scales and on a 25,000-prediction public network intrusion benchmark on deduplicated UNSW-NB15 flows [37]. Comparisons vary memory construction, inference context, refinement budget, and counterexample gating while tracking construction cost and rule quality.

The symbolic sanity check provides a known ground-truth rule with controllable irrelevant attributes (`texture` and `size`) and withheld combinations. Because this synthetic environment spans 16 combinations ($n = 48$ pooled evaluations across three masking seeds at Level 3), we use it strictly as a controlled sanity check to distinguish recovery of omitted evidence from improvements in the explanations themselves. It also exposes whether raw, abstract, or hybrid context best supports withheld combinations, and whether additional revision attempts justify their token cost.

The public-flow benchmark then tests the same memory-design questions on deduplicated 19-feature UNSW-NB15 telemetry across five 32-flow historical-memory seeds and 1000 shared test fingerprints, contextualized against shallow CART decision trees (depth 3 and unconstrained) as well as few-shot (32-flow) and full-train (88,699-flow) Random Forest and GBDT baselines. Specifically, the main contributions of this work are four-fold:

- **Auditable Local Consolidation and Counterexample Formulation:** We formulate VLEG, a parameter-free memory framework that enforces bounded cyclic generation–verification prior to memory admission and pairs accepted natural-language heuristics with raw episodic telemetry. We also formalize the theoretical boundary between single-episode source fidelity and cross-episode counterexample specificity.
- **Controlled Symbolic Sanity Check (7B and 70B):** Across five symbolic evaluations, we verify that bounded refinement ($K = 3$) eliminates source-label inversions (100% observed source-label agreement on admitted episodes versus 41.7%–50.0% for unverified proposals) and elevates admission coverage to 93.8%–100%, achieving accuracy comparable to raw retrieval at 70B (91.67% versus 89.58%, 44/48 versus 43/48) while confirming that single-episode verification cannot filter out coincidental distractor attributes.
- **Mechanistic Diagnosis of Reflection Collapse, Label Feedback, and False Alarms on UNSW-NB15:** Using a deterministic counterexample compiler that parses 99.4% (157/158) of admitted traffic rules into executable Boolean predicates, we show that unverified single-step reflection mislabels 95.0% (76/80) of malicious historical flows as benign (6.17% test attack recall). Crucially, analyzing the consolidation trajectory reveals that single-pass label feedback ($j = 1$) resolves only 14.5% (11/76) of these rejected malicious flows, whereas a second verification-gated revision ($j = 2$) recovers 63/76 (82.9%), proving the necessity of multi-step verification. We further show that standard VLEG trails raw retrieval (59.68% versus 63.92% Macro-F1) because 59.2% (93/157) of source-consistent rules match contrary-label flows within their 32-flow seed (72.0%, 113/157 across all 160 pooled flows).
- **Counterexample-Gated VLEG (VLEG-CG), Subset Disentanglement, and Decision-Tree / Ensemble Benchmarking:** Validating our gate on a disjoint 100-query pilot set, we show that within-seed counterexample gating (VLEG-CG and VLEG-CG-Mal) reduces the test false positive rate by 9.35–10.92 pp (from 48.78% to 37.86%–39.43%) and recovers overall Macro-F1 to 63.32%–63.55%. Disentangling gated versus un-gated queries reveals that on the un-gated subset where zero-counterexample rules are retained, VLEG-CG outperforms raw retrieval (64.60% versus 63.03% Macro-F1, −3.00 pp FPR). Finally, benchmarking

against shallow CART decision trees and RF/GBDT ensembles clarifies the exact trade-off between global axis-aligned split optimization and relational natural-language rule generation.

1.5. Paper Organization

The remainder of this paper moves from the memory-design choices motivating the study to the proposed framework and its empirical assessment. Section 2 reviews episodic retrieval, local reflection, semantic consolidation, and intrusion detection baselines, clarifying the differences in evidence retention, feedback, and inference context that guide our comparisons. Building on this discussion, Section 3 develops VLEG and VLEG-CG, explains the memory-consolidation and hybrid-inference phases, and specifies the verification loop, deterministic counterexample compiler, information-access boundaries, and overall algorithm. Section 4 then evaluates the framework through the 16-scene symbolic sanity check, the 25,000-prediction UNSW-NB15 benchmark, the iteration trajectory and subset audits, and the Decision Tree / Random Forest / GBDT comparisons, before discussing methodological limitations. Finally, Section 5 summarizes the principal conclusions and practical SOC deployment guidelines.

2. Related Work

2.1. Episodic Memory and Demonstration Retrieval

In-context learning uses demonstrations in the prompt to support task adaptation without parameter updates [1]. Retrieval-augmented generation (RAG) connects generation to external evidence retrieved for the current input [2]. LongMem extends access to past context through a memory bank and retrieval mechanism [10]. Related persistent-memory architectures including MemGPT [9], MemoryLLM [11], and HippoRAG [12] organize external experience across interaction sessions. Their memory contents and training procedures differ, so they should not be treated as interchangeable implementations of a single memory architecture.

Retrieval studies also address demonstration selection [20,21] and effective context length degradation [7,8,13], complementing broader surveys of agent memory design [14,15]. These questions concern which evidence reaches inference, independently of whether that evidence is subsequently summarized into verbal rules.

For labeled classification tasks, raw-example retrieval offers a direct, zero-consolidation-cost control: each historical record retains its observed features and ground-truth label. It can preserve local exceptions that a compressed summary omits, although its usefulness depends on selecting relevant examples within a finite context budget. A retrieved example also does not supply an explicit verbal rule or establish that a neighboring target has the same label. Retrieval relevance, evidence coverage, and prediction correctness therefore require separate evaluation.

Our raw-memory baseline follows this evidence-preserving principle. It provides labeled historical episodes at inference without reflection-generation calls. The symbolic sanity check compares it with abstract and hybrid context, while the public-flow experiment shares label-free feature retrieval across local conditions to control example selection. This design isolates the supplied memory content and gating mechanism more closely than a comparison that changes both retrieval and context construction.

2.2. Local Reflection and Feedback

MemoryBank stores, retrieves, and updates information from past interactions [22]. Generative Agents combine an experience store with higher-level synthesized reflections [23], while Reflexion converts binary task feedback into verbal reflections retained for subsequent trials [16]. These works show several ways to use experience beyond storing raw examples. They differ in what constitutes an episode, which feedback signal is available, when memories are updated, and how stored text influences later behavior.

Related work also examines procedural skill acquisition in open-ended worlds [17], structured agentic memory networks [24], and timeline-based memory management [25]. Their different memory objectives motivate explicit controls for the few-shot classification setting studied here.

The distinction between environmental feedback and source-episode verification is especially relevant. Trial feedback identifies an unsuccessful action or prediction and motivates revision across episodes. A source-episode check instead asks whether a candidate explanation agrees with the observed historical record under a specified verification prompt. Neither mechanism, by itself, proves that the resulting explanation identifies the true causal

factors across the broader data distribution. A reflection may repair its stated class label while retaining an unsupported threshold or treating an incidental attribute as a general condition.

VLEG studies a bounded per-episode admission procedure: generate a candidate from label-free attributes, check it against the labeled source, revise rejected candidates within budget K, and store an accepted reflection alongside its source evidence. Historical labels are available to the verifier, whereas target test labels are excluded from inference. By logging every intermediate revision attempt $j \in \{0, 1, 2\}$, our evaluation explicitly separates the effect of initial label disclosure in the first critique ($j = 1$) from multi-step verification enforcement ($j = 2$), as well as separating source-label agreement from cross-episode specificity.

2.3. Semantic Consolidation, Symbolic Rule Learners, and Intrusion Detection

Semantic consolidation compresses experience into global hypotheses or summaries that guide subsequent reasoning. Hypothesis-Driven Reasoning (HDR) iteratively generates and verifies hypotheses against episodic evidence, then combines hypotheses with retrieved episodes at inference [18]. It provides a close comparison for the unit of abstraction: hypotheses induced across multiple episodes versus VLEG's local reflections attached to individual source records. Global hypotheses can express conditional relations; indeed, one background-conditioned statement represents our synthetic sanity-check rule. Heterogeneity alone therefore does not imply that global abstraction must fail, though compressing diverse network attacks into a single global summary risks severe over-generalization.

Inductive reasoning and hypothesis generation are studied in [19, 26–29]; complementary work considers cognitive memory architectures [30–32], hallucination detection and verification [33, 34], and iterative self-critique [35,36]. In the domain of network traffic analysis and intrusion detection, supervised deep models such as ET-BERT [3], interpretable shallow decision trees (CART), and tabular ensemble classifiers (e.g., Random Forest and GBDT/XGBoost evaluated on UNSW-NB15 [37] and CICIDS2017 [4]) provide strong baselines. More recently, Houssel et al. [5] investigated LLMs for explainable network intrusion detection, and Islam et al. [6] proposed MA-IDS, a multi-agent RAG framework with an experience library for IoT intrusion detection.

Hybrid grounding supplies both an abstraction and supporting examples at inference. This retains access to concrete evidence, but can introduce conflicting guidance when the abstraction is inaccurate or applies too broadly. Its effectiveness depends on the specificity of the reflection, not merely the presence of both memory types. Comparing raw, abstract, hybrid, and counterexample-gated hybrid context against both interpretable decision trees and tree ensembles is therefore necessary to identify these operational trade-offs.

VLEG's contribution is a controlled diagnostic assessment of local admission checking, evidence–reflection pairing, and deterministic counterexample gating, contextualized directly against shallow CART decision trees and Random Forest / GBDT baselines under both 32-flow few-shot and 88,699-flow full-train regimes. The public-flow evaluation also includes an admission-matched control (`Raw on admitted flows`) that retains VLEG's admission mask while omitting reflections, strictly isolating reflection content from episode selection.

3. The Proposed Method

3.1. Overview of Verified Localized Episodic Grounding (VLEG)

VLEG stores a labeled historical episode together with an accepted local reflection, and screens admitted reflections against contrary-label historical counterexamples. The implementation distinguishes generation, source verification, counterexample compilation, and inference roles. The following formulation describes their information access and stopping behavior; source acceptance alone does not guarantee correctness of the verifier or universal validity of an accepted rule across unseen episodes.

Let $\mathcal{E} = \{e_1, \ldots, e_n\}$ be historical support episodes whose labels are available during consolidation. A pure episodic memory stores those labeled episodes directly ($\mathcal{M}_{\text{raw}} = \mathcal{E}$). The symbolic sanity check uses synthetic scene tuples, while the public-flow evaluation uses deduplicated UNSW-NB15 feature records from separate historical and test partitions. Write $e_i = (x_i, y_i)$, where x_i denotes observed feature attributes and y_i the historical class label. The initial generator call receives only x_i; the verifier receives (x_i, y_i), and its critique reveals y_i when a proposal is rejected. Test labels are strictly excluded from inference in all settings.

Conversely, global semantic consolidation [18] attempts to distill $\mathcal{E}$ into a compact set of universal rules or hypotheses, $\mathcal{H}$. This can be modeled as finding a global hypothesis set that maximizes the likelihood of the

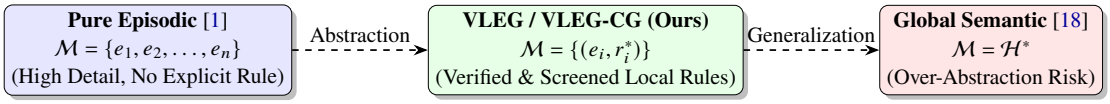


Figure 1. Theoretical positioning of VLEG and VLEG-CG within the memory abstraction spectrum. VLEG bridges pure episodic retention of raw traffic and global semantic consolidation by pairing raw episodes with source-verified and counterexample-screened local abstractions.

observed episodes:

$$\mathcal{H}^* = \arg\max_{\mathcal{H}} P(\mathcal{H} \mid e_1, e_2, \ldots, e_n). \tag{1}$$

A single conditional hypothesis can represent the contextual rule used in our symbolic sanity check. In complex network traffic datasets, however, a single global summary may omit important local exceptions or over-generalize attack conditions.

VLEG bridges this gap by shifting the abstraction target from the global distribution to the localized episode, while applying a model-based source-episode admission check (Figure 1). Rather than attempting to find a single universal $\mathcal{H}^*$, VLEG constructs a memory bank consisting of verified episodic–semantic tuples:

$$\mathcal{M}_{\text{VLEG}} = \left\{ (e_i, r_i^*) \;\middle|\; e_i \in \mathcal{E},\ \mathcal{V}(r_i^* \mid e_i) = 1 \right\}, \tag{2}$$

where r_i^* is a localized semantic reflection (a detection rule or heuristic) derived specifically from episode e_i, and $\mathcal{V}(\cdot \mid e_i)$ is a strict verification function that returns 1 when the implemented verifier accepts the candidate under its checking prompt.

The stored tuple retains the source episode alongside its accepted reflection. This permits later comparison with the concrete evidence, although the inference model may still follow an unsupported condition if the rule is overly broad.

3.2. Framework Components: Generator, Verifier, and Memory Bank

The consolidation architecture comprises a reflection generator, a strict source verifier, and a hybrid memory bank (Figure 2). Generation and checking use separate prompts and model calls to the same underlying foundation model, so shared model biases remain possible.

3.2.1. The Reflection Generator ($\mathcal{G}_\theta$)

The Reflection Generator is an LLM parameterized by θ, tasked with distilling a raw episode e_i into a localized natural-language abstraction r_i. Formally, at initial iteration $j = 0$, the generator samples a candidate reflection conditioned solely on the label-free attributes x_i:

$$r_i^{(0)} \sim \mathcal{G}_\theta(\cdot \mid x_i). \tag{3}$$

In subsequent iterations ($j > 0$), if the candidate reflection is rejected by the verifier, the generator is conditioned on the raw attributes x_i, its previous candidate reflection $r_i^{(j-1)}$, and a natural-language diagnostic critique $c_i^{(j-1)}$ provided by the verifier:

$$r_i^{(j)} \sim \mathcal{G}_\theta(\cdot \mid x_i, r_i^{(j-1)}, c_i^{(j-1)}). \tag{4}$$

The unverified single-step comparator stores its first proposal $r_i^{(0)}$ unconditionally without checking. Crucially, when $r_i^{(0)}$ is rejected due to a label mismatch, the critique $c_i^{(0)}$ explicitly informs the generator of the ground-truth label y_i; thus attempt $j = 1$ represents a label-conditioned revision, while attempt $j = 2$ represents a second verification-enforced revision if $r_i^{(1)}$ still fails verification.

3.2.2. The Strict Verifier ($\mathcal{V}_\phi$)

The verifier checks the candidate reflection against its single labeled historical episode $e_i = (x_i, y_i)$. It rejects mismatched class labels, fabricated feature values, or unparseable claims. The verifier outputs a tuple consisting of a binary validity score $v_i^{(j)} \in \{0, 1\}$ and a diagnostic critique $c_i^{(j)}$:

$$(v_i^{(j)}, c_i^{(j)}) = \mathcal{V}_\phi(e_i, r_i^{(j)}). \tag{5}$$

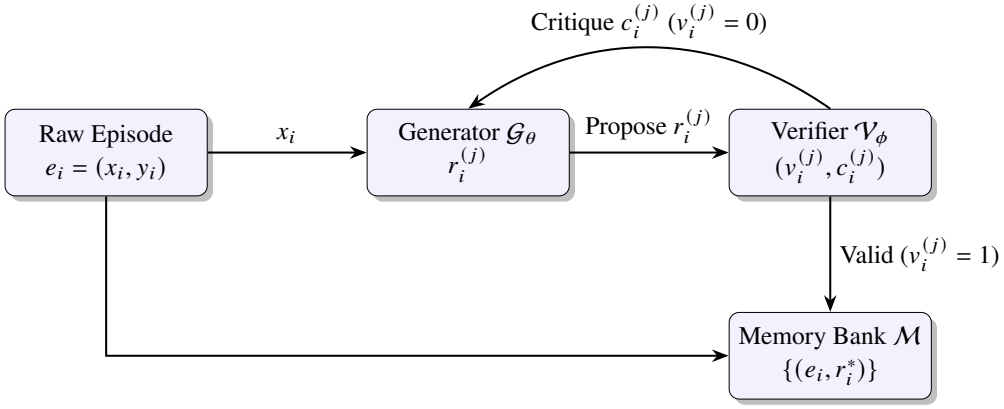


Figure 2. Architectural flow of the VLEG framework components. Separate generator and verifier calls admit only candidates accepted by the source check to the Hybrid Memory Bank alongside their raw episodic counterparts.

If $v_i^{(j)} = 1$, the candidate passes the admission check. If $v_i^{(j)} = 0$, its critique $c_i^{(j)}$ is returned to the generator. Admission establishes that the verifier accepted the reflection as consistent with episode e_i; as formalized in Section 3.4, single-episode acceptance does not by itself certify cross-episode specificity.

3.2.3. The Hybrid Memory Bank ($\mathcal{M}$)

Once a reflection passes the verifier at attempt j with $v_i^{(j)} = 1$, it is denoted as the verified reflection r_i^*. The Hybrid Memory Bank $\mathcal{M}$ stores the paired tuple of the raw episode and its verified abstraction:

$$\mathcal{M} = \bigcup_{i=1}^{n} \left\{ (e_i, r_i^*) \;\middle|\; \exists\, 0 \le j < K \text{ s.t. } \mathcal{V}_\phi(e_i, r_i^{(j)})_v = 1 \right\}. \tag{6}$$

Here K is the maximum number of generation–verification attempts, indexed $j = 0, \ldots, K-1$, including the initial proposal. The first accepted candidate is retained. If none is accepted within K attempts, the episode is omitted from $\mathcal{M}$. The memory bank supports retrieval of both source evidence and a local reflection in the hybrid modality, whereas raw-only and abstract-only ablations remove one of these components for controlled comparison.

3.3. Learning Mechanism: Iterative Cyclic Refinement

The refinement loop revises a rejected candidate using the verifier critique (Figure 3). It stops at the first accepted candidate or after K attempts. Reaching the budget ceiling without acceptance omits the episode from reflection memory.

Formally, let $e_i \in \mathcal{E}$ denote a raw historical episode. At any iteration $j \in \{0, \ldots, K-1\}$, letting $(r_i^{(-1)}, c_i^{(-1)}) = (\emptyset, \emptyset)$ at $j = 0$, the generator synthesizes a candidate reflection conditioned on the label-free attributes x_i and prior feedback:

$$r_i^{(j)} \sim \mathcal{G}_\theta(\cdot \mid x_i, r_i^{(j-1)}, c_i^{(j-1)}). \tag{7}$$

Once candidate $r_i^{(j)}$ is generated, the verifier $\mathcal{V}_\phi$ evaluates it against the full labeled source episode $e_i = (x_i, y_i)$ under a fail-closed native-Boolean JSON gate, outputting a binary validity decision $v_i^{(j)} \in \{0, 1\}$ and a diagnostic critique $c_i^{(j)}$:

$$(v_i^{(j)}, c_i^{(j)}) = \mathcal{V}_\phi\big((x_i, y_i),\ r_i^{(j)}\big). \tag{8}$$

Note that $\mathcal{V}_\phi$'s only argument besides the candidate reflection is the single episode e_i from which it was generated—by construction, the per-episode verifier has no access to any other episode in $\mathcal{E}_{-i} = \mathcal{E} \setminus \{e_i\}$. This check can expose a reflection that misstates e_i's own attributes or class label, but is not sufficient to determine whether an attribute the reflection treats as a sufficient condition is causally discriminative across episodes versus merely true of this single instance. This defines a fundamental boundary condition between single-episode source consistency and cross-episode specificity.

Define the stopping iteration j_i^* as the first attempt accepted by the verifier within the refinement budget K:

$$j_i^* = \min \left\{ 0 \le j < K \;\middle|\; v_i^{(j)} = 1 \right\}. \tag{9}$$

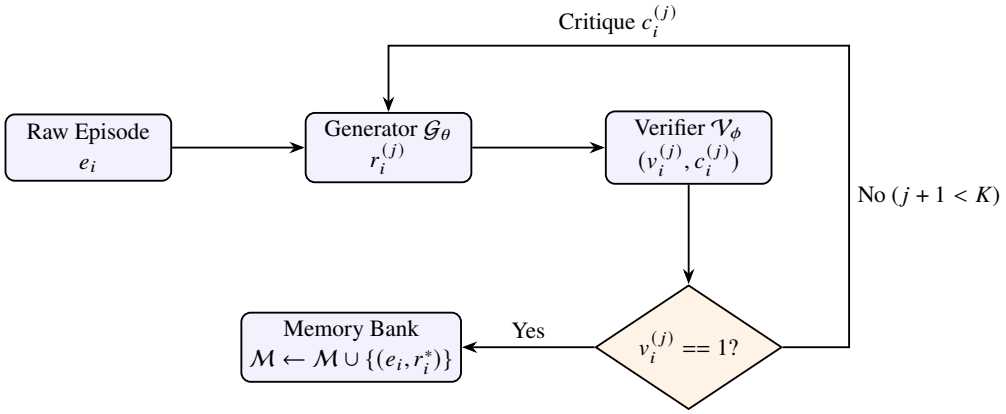


Figure 3. State transition diagram of the Iterative Cyclic Refinement mechanism. The loop stores the first candidate accepted by the verifier to the paired episodic–semantic memory bank.

If j_i^* exists, the loop terminates immediately at $j = j_i^*$ and commits $r_i^* = r_i^{(j_i^*)}$ to the memory bank; if $v_i^{(j)} = 0$ (including any malformed verifier response), the critique $c_i^{(j)}$ is fed back into the generator for iteration $j + 1$.

Remark 1 (Information-Access Scope of the Per-Episode Admission Check). Let A_j denote reaching attempt j, E_j^{local} a candidate reflection that contradicts its source episode $e_i = (x_i, y_i)$ (e.g., an inverted label or fabricated feature value), and B_j its erroneous acceptance by $\mathcal{V}_\phi$. Using one-based attempt indices $j = 1, \ldots, K$, if $\Pr(E_j^{\text{local}} \mid A_j) \le p_h$ and $\Pr(B_j \mid A_j, E_j^{\text{local}}) \le p_v$, the union bound over K attempts yields:

$$\Pr(\text{local source-contradiction commitment}) \le \sum_{j=1}^{K} \Pr(A_j \cap E_j^{\text{local}} \cap B_j) \le \min\{1, K p_h p_v\}. \tag{10}$$

Crucially, Equation (10) applies only to local source contradictions E_j^{local} detectable from e_i alone. For cross-episode over-generalization errors E_j^{cross}—where r_i^* is 100% consistent with e_i but also matches contrary-label episodes $e_m \in \mathcal{E}_{-i}$—a single-episode verifier has no discriminative signal ($\Pr(B_j \mid E_j^{\text{cross}}, \text{local agreement}) \approx 1$). Suppressing E_j^{cross} therefore requires an explicit cross-episode counterexample check across $\mathcal{E}_{-i}$.

3.4. Deterministic Counterexample Compiler and Cross-Episode Gating (VLEG-CG)

To diagnose and filter cross-episode over-generalization errors E_j^{cross} without incurring additional LLM API calls, we introduce a deterministic counterexample compiler Ψ that operates over admitted reflections in $\mathcal{M}$.

In our network traffic consolidation prompt, candidate reflections are structured as conjunctive conditional heuristics of the form `IF <condition_1> AND <condition_2> ... THEN <CLASS_LABEL>`. Given an admitted reflection r_i^*, the compiler Ψ parses each atomic clause into an executable relational predicate over the 19 flow features:

$$\pi(r_i^*)(x) = \bigwedge_{m=1}^{d_i} \text{op}_m\big(x[\text{field}_m],\ \text{rhs}_m\big) \in \{0, 1\}, \qquad \hat{y}(r_i^*) \in \{\texttt{BENIGN}, \texttt{MALICIOUS}\}, \tag{11}$$

where $\text{op}_m \in \{=, \neq, >, <, \ge, \le\}$ and rhs_m is either a numeric literal, a categorical string literal, or another feature column reference (such as `sbytes > dbytes`).

Definition 1 (Within-Seed and Cross-Seed Counterexample Counts and Gating Policies)**.** For any admitted tuple $(e_i, r_i^*) \in \mathcal{M}$ with a compilable predicate $\pi(r_i^*)$, let $\mathcal{E}_{\text{ref}}$ denote a reference set of historical episodes excluding e_i. The counterexample count $\text{CE}(r_i^* \mid \mathcal{E}_{\text{ref}})$ is the number of episodes in $\mathcal{E}_{\text{ref}}$ whose ground-truth label y_m contradicts the rule's predicted class $\hat{y}(r_i^*)$ when the rule antecedent fires:

$$\text{CE}(r_i^* \mid \mathcal{E}_{\text{ref}}) = \sum_{e_m = (x_m, y_m) \in \mathcal{E}_{\text{ref}}} \mathbb{I}\big(\pi(r_i^*)(x_m) = 1 \ \wedge \ y_m \neq \hat{y}(r_i^*)\big)\,. \tag{12}$$

We distinguish two reference scopes:

- **Within-Seed Counterexamples** ($\mathrm{CE}_{\mathrm{seed}}$, $\mathcal{E}_{\mathrm{ref}} = \mathcal{E}_{\mathrm{seed}} \setminus \{e_i\}$, 31 **flows):** Computed strictly within each seed's 32-flow support bank without accessing any other seed or test data. This quantity drives our online counterexample gates.
- **Cross-Seed Pooled Counterexamples** ($\mathrm{CE}_{\mathrm{pool}}$, $\mathcal{E}_{\mathrm{ref}} = \mathcal{E}_{\mathrm{all}} \setminus \{e_i\}$, 159 **flows):** Computed offline across the union of all 160 historical flows from the five seeds solely as a post-hoc diagnostic of broader population over-generalization.

When a target query T retrieves top-k memory tuples $\mathcal{M}_{\mathrm{top}}(T)$, we evaluate two primary online gating policies using the strict zero-counterexample threshold ($\mathrm{CE}_{\mathrm{seed}} = 0$, selected a priori and validated on the disjoint 100-query pilot partition):

1. **Malicious-Rule Counterexample Gate (VLEG-CG-Mal):** Because false alarms in SOC triage arise when overly broad malicious rules fire on normal traffic, VLEG-CG-Mal marks a tuple (e_i, r_i^*) as over-generalized if $y_i = \texttt{MALICIOUS}$ and $\mathrm{CE}_{\mathrm{seed}}(r_i^*) > 0$ (or if r_i^* is unparseable). If any tuple in $\mathcal{M}_{\mathrm{top}}(T)$ is marked, VLEG-CG-Mal strips the verbal rules from the prompt for query T and falls back to the raw admitted flow records $\{e_i\}_{(e_i, r_i^*) \in \mathcal{M}_{\mathrm{top}}(T)}$.
2. **All-Rule Counterexample Gate (VLEG-CG):** VLEG-CG applies the zero-counterexample requirement $\mathrm{CE}_{\mathrm{seed}}(r_i^*) = 0$ symmetrically across both benign and malicious rules in $\mathcal{M}_{\mathrm{top}}(T)$, falling back to raw admitted flow records whenever any retrieved rule matches a contrary-label historical episode within the seed's 32-flow support bank.

3.5. Inference Integration: Hybrid Episodic–Semantic Grounding

Once the memory bank $\mathcal{M}$ has been populated through Phase 1 consolidation, the system leverages this stored knowledge to classify a novel target query T. Traditional RAG and standard in-context learning rely on episodic-only retrieval [2, 20], injecting raw experiences directly into the context window. While raw retrieval preserves instance-level details, it provides no explicit, auditable decision rule at test time, requiring the inference model to perform implicit ad-hoc induction over high-dimensional telemetry.

To expose both representations at inference, VLEG uses *Hybrid Episodic–Semantic Grounding*, simultaneously injecting the raw telemetry e_i and its verified semantic rule r_i^* into the context window [14, 15]. Our empirical evaluations characterize the task-, scale-, and specificity-dependent boundary conditions where hybrid grounding yields advantages versus when counterexample gating is required to prevent false-positive inflation.

3.5.1. Domain-Aligned Retrieval

Given a novel target task T, the system retrieves the top-k most relevant memory tuples. The VLEG retrieval module supports two domain-aligned indexing modes: semantic reflection-embedding indexing for symbolic rules ($z_i = r_i^*$) and standardized feature-embedding indexing for tabular network flows ($z_i = x_i$, where rejected candidates are masked without replacement to strictly isolate memory-consolidation and gating effects from candidate-selection variance). Let $f_{\mathrm{emb}}(\cdot)$ denote a pre-trained dense embedding function. The relevance score $S(T, \mathcal{M}_i)$ for tuple $\mathcal{M}_i = (e_i, r_i^*)$ is computed via cosine similarity:

$$S(T, \mathcal{M}_i) = \frac{f_{\mathrm{emb}}(T) \cdot f_{\mathrm{emb}}(z_i)}{\|f_{\mathrm{emb}}(T)\| \|f_{\mathrm{emb}}(z_i)\|}. \tag{13}$$

The system then selects the subset of k tuples that maximize this similarity score:

$$\mathcal{M}_{\mathrm{top}} = \underset{\mathcal{S} \subseteq \mathcal{M},\, |\mathcal{S}| = \min(k, |\mathcal{M}|)}{\arg\max} \sum_{(e_i, r_i^*) \in \mathcal{S}} S(T, (e_i, r_i^*)). \tag{14}$$

3.5.2. Hybrid and Counterexample-Gated Prompt Formulation

After retrieving $\mathcal{M}_{\mathrm{top}}$, the inference module constructs the context window. Under standard VLEG (and un-gated queries under VLEG-CG), the verified reflection r_i^* provides local heuristic guidance while the labeled source

Algorithm 1 Verified Localized Episodic Grounding with Counterexample Gating (VLEG / VLEG-CG)

Input: Episodes $\mathcal{E}_{\text{seed}}$, target T, max attempts K, retrieval count k, retriever R, gate mode Gate $\in$ {None, CG-Mal, CG-All}

Initialize Memory Bank $\mathcal{M} \leftarrow \emptyset$
// Phase 1: Memory Consolidation and Within-Seed Counterexample Compilation
for each $e_i = (x_i, y_i) \in \mathcal{E}_{\text{seed}}$ **do**
  $j \leftarrow 0$; $(r, c) \leftarrow (\emptyset, \emptyset)$; $v \leftarrow 0$
  while $v = 0$ **and** $j < K$ **do**
    $r_i^{(j)} \leftarrow \mathcal{G}_\theta(x_i, r, c)$ ▷ Generate or refine candidate reflection
    $(v, c) \leftarrow \mathcal{V}_\phi(e_i, r_i^{(j)})$ ▷ Check reflection against labeled source e_i
    $r \leftarrow r_i^{(j)}$; $j \leftarrow j + 1$
  end while
  if $v = 1$ **then**
    $r_i^* \leftarrow r_i^{(j-1)}$
    $\text{CE}_i \leftarrow \text{CE}(r_i^* \mid \mathcal{E}_{\text{seed}} \setminus \{e_i\})$ via compiler Ψ (Equation (12))
    $\mathcal{M} \leftarrow \mathcal{M} \cup \{(e_i, r_i^*, \text{CE}_i)\}$ ▷ Store verified tuple and within-seed CE count
  end if
end for
// Phase 2: Hybrid Episodic–Semantic Inference
$\mathcal{M}_{\text{top}} \leftarrow R(\mathcal{M}, T, k)$ ▷ Retrieve top-k relevant tuples
if Gate $\neq$ None **and** $\mathcal{M}_{\text{top}}$ violates counterexample gate **then**
  Context $\leftarrow$ PromptBuilder($\{(e_i, \emptyset)\}_{(e_i, r_i^*) \in \mathcal{M}_{\text{top}}}, T$) ▷ Fallback to raw admitted flows
else
  Context $\leftarrow$ PromptBuilder($\mathcal{M}_{\text{top}}, T$) ▷ Construct hybrid episode + rule prompt
end if
Output $\leftarrow I_\psi$(Context) ▷ Generate final classification
return Output

episode e_i provides concrete evidence:

$$Q_i = \begin{cases} \big[\text{Episode: } e_i\big], & \text{if gated by VLEG-CG / VLEG-CG-Mal,} \\ \big[\text{Episode: } e_i \oplus \text{Verified Rule: } r_i^*\big], & \text{otherwise,} \end{cases}$$
$$P = \text{PromptBuilder}\Big(\bigoplus_{(e_i, r_i^*) \in \mathcal{M}_{\text{top}}} Q_i \oplus \text{Target: } T \Big), \tag{15}$$

where $\oplus$ denotes sequence concatenation. The reasoning LLM, parameterized by ψ, then generates the final prediction $\hat{y}$ conditioned on prompt P:

$$\hat{y} = \text{LLM}_\psi(P). \tag{16}$$

3.6. Final Algorithm and Execution Flow

The VLEG and VLEG-CG framework synthesizes the Reflection Generator $\mathcal{G}_\theta$, Strict Verifier $\mathcal{V}_\phi$, Counterexample Compiler Ψ, and Hybrid Inference Module I_ψ into a cohesive two-phase pipeline (Algorithm 1): **Phase 1: Memory Consolidation and Counterexample Screening** and **Phase 2: Hybrid Inference**.

During Phase 1, the system ingests historical episodes $\mathcal{E} = \{e_1, e_2, \ldots, e_n\}$. For each episode e_i, the Generator $\mathcal{G}_\theta$ proposes $r_i^{(0)}$ from label-free attributes x_i. The Verifier $\mathcal{V}_\phi$ evaluates this candidate against the grounding episode $e_i = (x_i, y_i)$, outputting binary validity flag $v \in \{0, 1\}$ and critique c. If $v = 0$, critique c is fed back into $\mathcal{G}_\theta$ to produce a refined reflection $r_i^{(j)}$ until $v = 1$ or budget K is reached. For each admitted reflection r_i^*, compiler Ψ computes its within-seed counterexample count $\text{CE}_{\text{seed}}(r_i^*) = \text{CE}(r_i^* \mid \mathcal{E}_{\text{seed}} \setminus \{e_i\})$ via Equation (12).

During Phase 2, given a novel target query T, Retriever R extracts the top-k most relevant memories $\mathcal{M}_{\text{top}} \subset \mathcal{M}$. If counterexample gating is active and a retrieved rule violates the zero-counterexample condition, the prompt builder strips the over-generalized verbal rules and presents the raw admitted episodes; otherwise, the full hybrid tuples (e_i, r_i^*) are supplied to I_ψ to generate the final classification.

4. Experiments

4.1. Experimental Settings and Two-Stage Protocol

Our experimental evaluation is organized into two stages: (1) a controlled 16-scene symbolic sanity check (Sections 4.2–4.6) designed to verify the mechanics of per-episode verification, grounding modalities, refinement budgets, and distractor attribution against a known oracle, and (2) a large-scale public network-flow benchmark on deduplicated UNSW-NB15 telemetry (Sections 4.7–4.8) comprising 25,000 formal LLM predictions, iteration-level consolidation audits, disjoint pilot threshold validation, un-gated vs. gated subset disentanglement, and direct comparisons against shallow CART decision trees and Random Forest / GBDT ensembles.

The symbolic sanity check uses synthetic scenes defined by shape, color, texture, background, and a deterministic size distractor. On white backgrounds, `ANOMALY` requires a red circle; black backgrounds invert this rule. Texture and size are non-causal distractors. Level 1 contains eight white-background combinations; Level 2 contains all sixteen combinations across both backgrounds. Both evaluate combinations used to form memory. Level 3 withholds four combinations from the 12-episode support bank and tests all sixteen; masked-only accuracy measures withheld-combination transfer. Masking seeds change the withheld subset rather than the underlying 16-scene population.

Symbolic runs use Qwen2.5-7B-Instruct or Llama-3.3-70B-Instruct, `nomic-embed-text` embeddings, and temperature 0. Generator and verifier are separate calls to the same model. Initial generation excludes historical labels; verification receives labeled sources, and later attempts can receive labels through supervised critique. Target labels are reserved for scoring. We vary consolidation strategy, grounding modality, and $K \in \{1, 2, 3, 5\}$, measuring accuracy, memory coverage, source-label consistency, distractor attribution flags, and construction tokens. Cross-scale comparisons between 7B (Qwen2.5) and 70B (Llama-3.3) also change model family and therefore cannot isolate parameter count causally.

4.2. Controlled Memory-Strategy Comparison (Symbolic Sanity Check)

On the 16-scene symbolic sanity check, verified local consolidation eliminates the label-inversion collapse of unverified reflection and reaches accuracy comparable to raw retrieval at 70B (91.67% **versus** 89.58%**,** 44/48 **versus** 43/48**), while trailing raw retrieval at 7B (**75.0% **versus** 87.5%**).** To isolate core consolidation mechanisms under identical model weights and retrieval budgets, we compare VLEG ($K = 3$) against global-hypothesis consolidation (HDR adaptation), unverified single-step reflection, and raw in-context learning (ICL) across the three symbolic complexity levels.

Within each model-scale comparison, the same underlying model and temperature (0) are used. Level 1 has eight queries and Level 2 has sixteen. Level 3 is pooled over three masking seeds, giving 48 evaluations of the sixteen recurring scene combinations. Repeat runs matched reported accuracies in the checked conditions, representing an empirical repeatability observation under greedy decoding.

As shown in Figure 4, the 7B accuracies for raw retrieval (ICL) are 100.0%, 93.8%, and 87.5% at Levels 1–3, compared with VLEG at 75.0%, 87.5%, and 75.0%. Global-hypothesis consolidation (HDR) recovers the uniform-background rule at Level 1 (100.0%), but drops to 62.5% at Level 2 and a pooled accuracy of 70.8% at Level 3 once the rule becomes background-conditional. The unverified single-step reflection comparator scores 25.0%, 50.0%, and 50.0%; inspected first proposals predominantly state `ANOMALY`, so its aggregate accuracy tracks the anomaly base rate rather than discriminating between scenes.

At the 7B scale, VLEG outperforms unverified single-step reflection by up to +50.0 pp and exceeds global hypothesis consolidation on background-conditional rules by +25.0 pp at Level 2 and +4.2 pp at Level 3, while trailing raw retrieval.

The 70B evaluation tests Llama-3.3-70B-Instruct across the three Level-3 masking seeds ($n = 48$) and pairs VLEG against a stronger multi-attempt binary-feedback Reflexion comparator as well as ICL and HDR. At 70B, Level-3 pooled accuracies are VLEG 91.67% (44/48), raw retrieval (ICL) 89.58% (43/48), global hypothesis (HDR) 83.33% (40/48), and feedback-based Reflexion 81.25% (39/48). VLEG and raw retrieval perform comparably at 70B (differing by one query out of 48, with VLEG higher in two seeds and lower in the third). Because the model family also changes between 7B and 70B and the symbolic population is small, this shift should be interpreted as a model-dependent capability difference rather than an isolated scaling law.

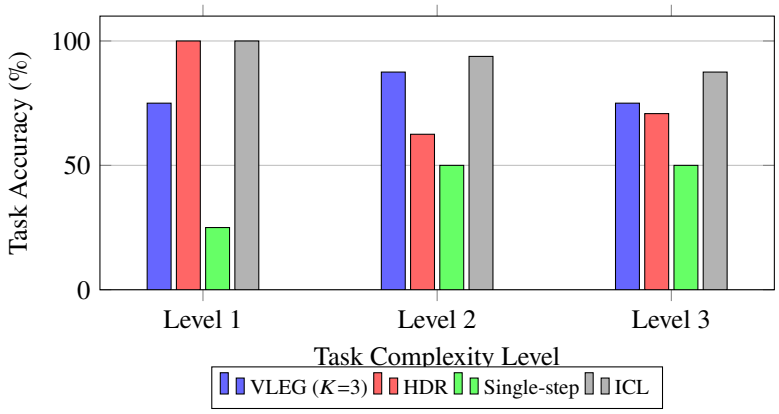


Figure 4. Measured Task Accuracy (%) across three levels of symbolic task complexity (Qwen2.5-7B-Instruct; Level 3 pooled across 3 masking seeds, $n = 48$). ICL leads at every level at 7B; HDR is bimodal (perfect at Level 1's non-conditional rule, weaker once the rule becomes background-conditional); unverified Single-step tracks each level's `ANOMALY` base rate.

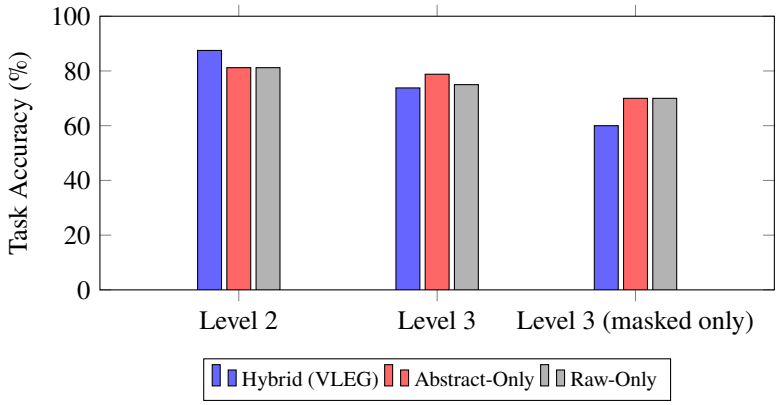


Figure 5. Measured Task Accuracy across grounding modalities (Qwen2.5-7B-Instruct). Level 3 figures are pooled across 5 masking seeds ($n = 80$ overall, $n = 20$ for the masked-only subset). Hybrid Grounding leads on Level 2 (87.5%) but trails single-modality baselines on unseen Level-3 combinations at 7B (60.0% versus 70.0%).

4.3. Component Isolation (Grounding Modality Ablation)

Pairing verified reflections with raw episodes improves seen-rule classification at 7B (87.5%) but penalizes unseen combinations (60.0% versus 70.0%) due to distractor-attribute anchoring, a penalty that narrows at the 70B scale. To isolate inference-context composition from retrieval selection, this ablation fixes the retrieved top-k tuples using reflection-embedding similarity while varying whether the prompt displays both (e_i, r_i^*) in Hybrid Grounding, only r_i^* in Abstract-Only, or only e_i in Raw-Only [1].

At the 7B scale, Figure 5 shows that Hybrid Grounding achieves the highest accuracy on seen combinations at Level 2 (87.5% versus 81.2% for both single-modality baselines). However, on Level 3 ($n = 80$ pooled evaluations across five masks), Abstract-Only exceeds Hybrid by 5.0 pp overall (78.8% versus 73.8%, with Raw-Only at 75.0%) and by 10.0 pp on withheld masked combinations (70.0% versus 60.0%, $n = 20$). Hybrid does not outperform both alternatives in any individual 7B Level-3 seed.

At the 70B scale across the same five masks, the overall ordering reverses in favor of evidence-retaining modalities, with Hybrid and Raw-Only tying at 95.0% overall compared with 91.25% for Abstract-Only, while Abstract-Only retains a narrower 5.0 pp edge on the masked subset (85.0% versus 80.0%, with per-seed masked accuracies of 100%, 75%, 50%, 100%, and 100%).

These results separate overall performance from withheld-combination transfer. Supplying raw evidence alongside a reflection does not guarantee a benefit when the reflection embeds incidental distractor attributes (`texture` and `size`) that happen to hold in the source episode. When both e_i and r_i^* are presented for an unseen Level-3 combination, the 7B model can over-anchor on spurious conjunctions shared between the raw example and the rule text, whereas the 70B model generates fewer distractor references during consolidation (Section 4.6).

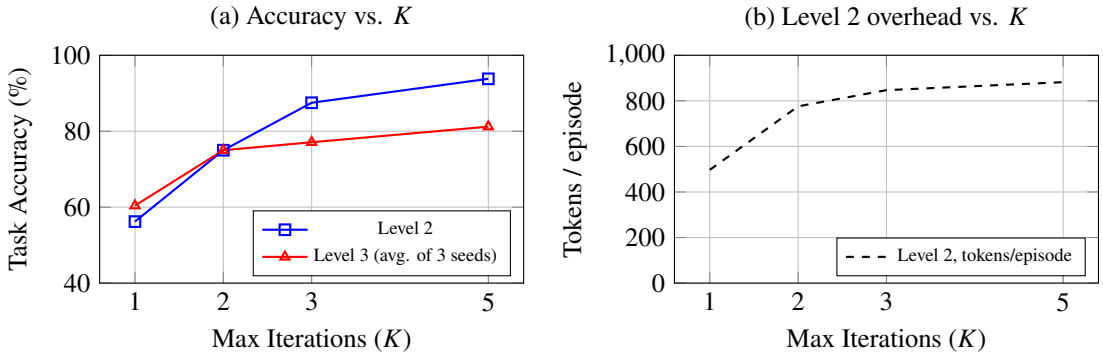


Figure 6. Measured parameter sensitivity of the maximum refinement iterations K (Qwen2.5-7B-Instruct). (a) Task Accuracy continues improving through $K = 5$ at 7B; (b) token overhead per consolidated episode is concave, with most of the cost increase occurring between $K = 1$ and $K = 2$.

4.4. Iterative Refinement Ablation

Bounded cyclic refinement ($K = 3$) substantially improves downstream accuracy primarily by recovering rejected historical episodes rather than by purifying causal explanations. Both $K = 1$ and $K = 3$ employ the same generator, strict verifier, and hybrid inference module, stopping at the first accepted proposal. Whereas $K = 1$ immediately discards any episode whose initial reflection fails verification, $K = 3$ allows up to two critique-guided revisions before omission.

Across five 7B masking seeds, $K = 3$ achieves 75.0% overall accuracy (60/80), compared with 63.75% (51/80) at $K = 1$, an 11.25 pp increase. Masked-only accuracy rises from 50.0% (10/20) to 60.0% (12/20). The masked-only difference is zero in three seeds and positive in two. The pooled overall gain therefore does not imply that every memory draw improves transfer to withheld combinations. These are repeated evaluations of the same sixteen-element population under different masks, not eighty independent scenes.

The principal observed change is memory admission coverage. First-attempt verification ($K = 1$) retains 42%–58% of historical episodes across seeds, whereas $K = 3$ retains 83%–100%. Rejection critique can reveal the correct historical label, allowing a revised candidate to agree with its source. Additional attempts consequently recover examples that the smaller budget would omit. The downstream improvement is consistent with this coverage recovery pathway; it is not evidence that each recovered explanation identifies the true causal attributes more reliably.

In the 70B comparison across the same five masks, increasing the budget from $K = 1$ to $K = 3$ yields a mean overall accuracy gain of 26.2 ± 8.1 pp, admitting all twelve historical episodes (100%) at $K = 3$ in every seed. Neither the pooled gain nor full admission proves universal causal generalization outside the synthetic oracle.

The comparison clarifies what refinement can and cannot change. Increasing K changes the opportunity to revise and the resulting evidence available at inference; it does not add new source observations or independently validate a generalized rule across episodes. Coverage recovery and explanation specificity must therefore be evaluated as distinct dimensions, since an accepted reflection can match its historical label while still retaining an incidental attribute. Overall, these findings validate bounded refinement as an effective mechanism for preventing rejection-induced memory starvation while motivating a joint analysis of accuracy gains against token construction overhead.

4.5. Parameter Sensitivity and Overhead Analysis

Consolidation token overhead grows concavely with the refinement ceiling K because most historical episodes converge within two verification attempts. To evaluate the efficiency frontier of cyclic refinement, we sweep the maximum budget over $K \in \{1, 2, 3, 5\}$ and track both task accuracy and total prompt-plus-completion tokens per consolidated episode. At $K = 1$, verification still occurs, but a rejected proposal cannot be revised. Level 2 uses the fixed 16-scene population; Level 3 varies the withheld subset across three masking seeds.

Figure 6(a) shows no uniform 7B plateau at $K = 3$: Level-2 accuracy rises from 87.5% at $K = 3$ to 93.8% at $K = 5$, and Level-3 mean accuracy rises from 77.1% to 81.2% (with only one of the three individual masks plateauing at $K = 3$). A budget sufficient for one memory draw can therefore remain insufficient for another at

Table 1. Diagnostic assessment of memory quality on the symbolic sanity check (Qwen2.5-7B-Instruct). Admission is stored episodes divided by historical episodes. Label correctness and spurious-attribution rate are computed deterministically against ground truth. "Spurious (of correct)" = % of label-correct reflections containing a texture/size keyword.

Level	Strategy	Admission	Label-Correct	Spurious (of correct)
Level 2	Unverified Single-Step	N/A (Always commit)	50.0% (8/16)	87.5% (7/8)
	VLEG (K=3)	**93.8% (15/16)**	**100.0% (15/15)**	86.7% (13/15)
Level 3	Unverified Single-Step	N/A (Always commit)	41.7% (5/12)	80.0% (4/5)
	VLEG (K=3)	**100.0% (12/12)**	**100.0% (12/12)**	83.3% (10/12)

7B.

As shown in Figure 6(b), construction overhead grows concavely over the measured ceilings, rising by 279 tokens per episode from $K = 1$ to $K = 2$ (497 to 776 tokens), by 71 tokens from $K = 2$ to $K = 3$ (847 tokens), and by only 35 tokens from $K = 3$ to $K = 5$ (882 tokens). Because the vast majority of episodes pass verification on the first or second attempt, extending K invokes additional LLM calls only for the small tail of unresolved episodes.

At 70B, all three Level-3 masks yield identical admitted memories, token totals, and accuracy across $K = 2$, $K = 3$, and $K = 5$, as no historical episode requires a third attempt in those runs. The observed convergence point is therefore $K = 2$ for the 70B model on this sanity check, rather than a universal stopping rule across scales. Wall-clock timing is not used to rank these historical runs because concurrent GPU workloads affected latency, and a Level-2 repeat at $K = 3$ shifted total tokens slightly from 12,541 to 12,625 while preserving accuracy, confirming that temperature 0 does not guarantee bit-exact hardware determinism.

4.6. Diagnostic Assessment of Memory Quality

Single-episode verification enforces 100% observed source-label agreement on admitted symbolic memories across scales, whereas suppressing non-causal distractor attributes depends on foundation-model characteristics rather than local admission checking. To disentangle what per-episode verification filters from what passes through, we evaluate admitted-memory coverage, deterministic source-label consistency, and keyword references to non-causal attributes (`texture` and `size`).

Coverage is the proportion of historical episodes admitted within $K = 3$; it is not applicable as a filtering metric for an always-commit comparator. Source-label consistency is scored deterministically against the ground-truth scene label. The spurious-attribution flag detects references to `texture` or `size`, which have zero effect on the ground-truth label. Because a keyword flag detects any mention as well as a causal assertion, it serves as a conservative upper-bound diagnostic.

Table 1 reports 100.0% observed source-label agreement among admitted VLEG reflections at both levels (15/15 at Level 2 and 12/12 at Level 3), compared with 41.7%–50.0% for the unverified single-step comparator. Conditional on label correctness, however, unsupported-attribution rates are 86.7% versus 87.5% at Level 2 and 83.3% versus 80.0% at Level 3, showing no verification-specific reduction in distractor mentions.

In this diagnostic, admitted reflections match their source labels but frequently cite attributes that do not govern the synthetic rule. For example, asserting that a striped texture is required for an anomaly cannot be falsified by inspecting a single striped anomalous scene. Cross-episode contrast is required to separate an incidental instance property from a necessary class condition.

At 70B (Llama-3.3-70B-Instruct), aligning admitted reflections with their originating support episodes likewise yields 100.0% source-label consistency, compared with 50.0% for unverified first proposals. Unconditional texture/size keyword rates at 70B drop to 12.5% (2/16) at Level 2 and 16.7% (2/12) at Level 3 for both VLEG and unverified proposals alike (down from 83.3%–93.8% at 7B). Because this reduction occurs equally in both strategies, it reflects the underlying model's preference for parsimonious rules rather than a filtering effect of single-episode verification.

This distinction is central to interpreting any verified memory bank: admission is conditioned on a single source episode, whereas downstream classification depends on cross-episode generalization. Agreement at the first stage cannot certify specificity at the second.

Table 2. Public-flow classification performance on the 1000 shared deduplicated UNSW-NB15 test fingerprints (mean ± sample standard deviation across five historical-memory seeds, in percent). Invalid LLM outputs count as errors. All few-shot methods use the exact same 32 historical flows per seed (16 benign, 16 malicious). Full-train supervised baselines are trained on all 88,699 deduplicated training flows across the 5 random seeds.

Supervision Regime	Condition / Method	Accuracy (%)	Macro-F1 (%)	Attack Recall (%)	False Positive Rate (%)
Few-Shot LLM Memory (32 flows/seed, k=3, Llama-3.3-70B, 0-train)	Raw flow retrieval	**65.08 ± 2.93**	**63.92 ± 2.54**	71.39 ± 8.44	38.13 ± 6.85
	Raw on admitted flows (control)	64.86 ± 2.96	63.70 ± 2.68	71.16 ± 8.71	38.34 ± 6.61
	Unverified single-step reflection	61.38 ± 1.08	42.66 ± 1.35	6.17 ± 1.43	**9.89 ± 2.03**
	Global summary	50.22 ± 12.45	47.60 ± 14.13	**87.12 ± 22.27**	68.54 ± 24.20
	VLEG (source-verified only, K=3)	59.74 ± 2.03	59.68 ± 2.03	77.74 ± 5.69	48.78 ± 1.67
	VLEG-CG-Mal (Ours, malicious CE gate)	64.16 ± 1.92	63.32 ± 2.10	72.11 ± 8.20	39.43 ± 4.68
	VLEG-CG (Ours, all-rule CE gate)	64.66 ± 2.83	63.55 ± 2.49	70.27 ± 8.14	**37.86 ± 7.16**
Few-Shot Tabular ML (Same 32 flows/seed, 5-seed mean ± SD)	Shallow Decision Tree (CART, depth=3)	70.18 ± 1.64	69.39 ± 1.76	83.38 ± 12.86	**36.53 ± 6.76**
	Decision Tree (CART, unconstrained)	**70.80 ± 0.90**	**70.25 ± 0.64**	86.41 ± 8.12	37.13 ± 5.08
	Random Forest (200 trees, 32-flow)	68.86 ± 1.25	68.42 ± 1.23	**87.60 ± 10.86**	40.66 ± 5.66
	Gradient Boosting / GBDT (32-flow)	66.60 ± 4.46	65.84 ± 4.60	78.04 ± 10.27	39.22 ± 5.52
Full-Train Tabular ML (88,699 unique flows, 5-seed mean ± SD)	Shallow Decision Tree (CART, depth=5)	77.20 ± 0.00	76.81 ± 0.00	95.25 ± 0.00	31.98 ± 0.00
	Random Forest (Full-train upper bound)	**82.06 ± 0.36**	**81.31 ± 0.36**	91.99 ± 0.59	**22.99 ± 0.55**
	Gradient Boosting / GBDT (Full-train)	76.50 ± 0.00	76.25 ± 0.00	**98.22 ± 0.00**	34.54 ± 0.00

Table 3. Deterministic counterexample compiler (Ψ) and iteration-level consolidation audit across the five historical-memory seeds (160 total historical episodes: 80 benign, 80 malicious). We report both within-seed counterexamples ($CE_{seed} > 0$ across the 31 other flows in the same seed, used by the online gate) and cross-seed pooled counterexamples ($CE_{pool} > 0$ across all 159 other historical flows, used for offline diagnosis).

Consolidation Stage / Method	Admitted	Benign Match	Malicious Match	Parseable	Within-Seed CE (All / Mal)	Mean CE_{seed} (Mal)	Cross-Seed CE (All / Mal)
Attempt j=0 (Unverified single-step)	160 / 160	80 / 80 (100%)	**4 / 80 (5.0%)**	160 / 160	126/160 (78.8%) / 66/80 (82.5%)	7.58	136/160 (85.0%) / 73/80 (91.3%)
Attempt $j \leq 1$ (1st label-informed revision)	95 / 160	80 / 80 (100%)	**15 / 80 (18.8%)**	—	—	—	—
Attempt $j \leq 2$ (VLEG K=3 final)	**158 / 160**	**80 / 80 (100%)**	**78 / 78 (100%)**	**157 / 158**	**93/157 (59.2%) / 33/77 (42.9%)**	**1.74**	**113/157 (72.0%) / 50/77 (64.9%)**

4.7. Public-Flow Benchmark on UNSW-NB15: Counterexample Audit, Pilot Selection, and Subset Analysis

On deduplicated UNSW-NB15 telemetry, standard VLEG resolves the recall collapse of unverified reflection (+17.02 **pp Macro-F1) via multi-step verification, while within-seed counterexample gating (VLEG-CG) reduces the false positive rate by** 9.35–10.92 **pp, recovers overall Macro-F1 to** 63.32%–63.55%**, and outperforms raw retrieval** (64.60% **versus** 63.03% **Macro-F1) on the un-gated subset where zero-counterexample rules are retained.** We evaluate public UNSW-NB15 testbed flows [37] following deterministic 19-feature deduplication, conflicting-label exclusion, and strict historical/test partition separation. Specifically, out of 175,341 raw training rows, our preparation pipeline removes 6896 rows belonging to 273 conflicting-label feature groups and 79,746 duplicate feature rows (45.5%), retaining 88,699 unique training flow fingerprints. Out of 82,332 raw testing rows, the pipeline removes 2912 conflicting rows (105 groups), 33,161 within-test duplicate rows (40.3%), and 1044 cross-partition train/test duplicates, leaving 45,215 disjoint test fingerprints. Five historical-memory seeds (`20261020`–`20261024`) each sample 32 historical records (16 benign and 16 malicious; 160 total historical episodes across the five seeds) and evaluate the exact same 1000 formal test fingerprints (663 benign, 337 malicious), which are strictly disjoint from the 100-query pilot partition (`pilot_20261012`). The dual-A800 server runs Llama-3.3-70B-Instruct (`Q4_K_M`), `nomic-embed-text` embeddings, temperature 0, $K = k = 3$, and a 32,768-token context window.

Raw flow retrieval, unverified single-step reflection, standard VLEG ($K = 3$), the admission-matched control (`Raw on admitted flows`), and our counterexample-gated variants (**VLEG-CG-Mal** and **VLEG-CG**) all share identical label-free feature-embedding retrieval ($z_i = x_i$); rejected candidates are masked without replacement. The admission control retains VLEG's exact admission mask (158/160 episodes) but omits verbal reflections, strictly isolating the effect of the injected reflection text from episode selection. A fifth LLM baseline conditions inference on a single consolidated `Global summary`.

4.7.1. Why Unverified Reflection Collapses and Why Single-Pass Label Disclosure is Insufficient

As shown in Table 2 and Table 3, unverified single-step reflection suffers a severe sensitivity drop on UNSW-NB15, achieving only 6.17% ± 1.43% attack recall and 42.66% ± 1.35% Macro-F1 (barely above an always-benign predictor at 39.87% Macro-F1). Table 3 (Row 1) reveals the exact cause: when generating candidate reflections from label-free flow attributes at $j = 0$, the LLM correctly labels 80/80 (100.0%) benign flows as `BENIGN`, but

mislabels 76/80 (95.0%) of malicious historical flows as `BENIGN` (identifying only 4/80 as `MALICIOUS`).

A critical methodological question is whether VLEG's improvement over unverified single-step reflection stems merely from receiving the ground-truth label in the critique rather than from iterative verification gating. Inspecting the logged per-iteration verification histories across all 160 support episodes (Table 3, Row 2) resolves this question directly. When the 76 mislabeled malicious episodes fail verification at $j = 0$, the verifier's critique explicitly reveals the ground-truth label (`stated label (BENIGN) contradicts historical label (MALICIOUS)`). Yet at attempt $j = 1$—which is conditioned directly on this ground-truth label feedback—**only 11 of the 76 (14.5%) malicious episodes are successfully revised to `MALICIOUS`**, whereas the remaining **65/76 (85.5%) still output `BENIGN` despite the label hint** and are rejected a second time by the verifier. Only at attempt $j = 2$ (the second verification-gated revision) do 63 of those 65 episodes finally pass verification (with 2/80 remaining rejected at $K = 3$). Thus, single-pass label disclosure without a subsequent verification gate ($j \leq 1$) leaves 65/80 (81.25%) of malicious flows with inverted labels, whereas iterative verification gating ($j = 2$) accounts for 63/78 (80.8%) of all admitted malicious rules.

4.7.2. Within-Seed vs. Cross-Seed Counterexample Diagnosis and Pilot Threshold Validation

Enforcing VLEG ($K = 3$) restores 100.0% source-label agreement across all 158/160 admitted episodes (80/80 benign, 78/78 malicious), raising test Macro-F1 by +17.02 pp over single-step reflection (to 59.68%, paired 95% CI $[13.71, 20.39]$ pp) and attack recall to 77.74%. However, standard VLEG trails zero-cost raw flow retrieval (63.92% Macro-F1) by −4.24 pp (paired 95% CI $[-5.91, -2.53]$ pp) because its FPR rises from 38.13% to 48.78% (+10.65 pp). Because the admission-matched control (`Raw on admitted flows`, 63.70% Macro-F1, 38.34% FPR) matches full Raw retrieval, this +10.65 pp FPR inflation is caused directly by the injected reflection rules.

Table 3 (Row 3) quantifies rule over-generalization under both counterexample scopes defined in Equation (12). Across the 157/158 (99.4%) parseable VLEG rules, 59.2% **(93/157) match at least one contrary-label flow within their own 32-flow seed** ($CE_{seed} > 0$, comprising 33/77 = 42.9% of malicious rules with a mean of 1.74 within-seed CEs, and 60/80 = 75.0% of benign rules). When evaluated offline across all 160 pooled historical flows ($CE_{pool} > 0$), 72.0% **(113/157) match at least one contrary-label flow across seeds** (50/77 = 64.9% of malicious rules with a mean of 8.16 cross-seed CEs, and 63/80 = 78.8% of benign rules).

To ensure that our online gating threshold ($CE_{seed} > 0$) involves no test-set leakage, we validated candidate thresholds $CE_{seed} > \tau$ for $\tau \in \{0, 1, 2\}$ on the disjoint 100-query pilot partition (`pilot_20261012`). On the pilot set, unscreened VLEG scored 48.87% Macro-F1 (61.43% FPR) compared with 54.63% Macro-F1 (54.29% FPR) for Raw retrieval. Setting the strict zero-counterexample threshold $\tau = 0$ achieved the best pilot Macro-F1 (54.63% for VLEG-CG and 54.45% for VLEG-CG-Mal) and lowest pilot FPR (52.86%–54.29%), outperforming looser thresholds $\tau = 1$ (51.69%–52.77% Macro-F1, 57.14% FPR) and $\tau = 2$ (48.87%–53.83% Macro-F1, 57.14%–61.43% FPR).

4.7.3. Un-Gated vs. Gated Subset Analysis: Do Screened Rules Help When Retained?

On the 1000 formal test fingerprints across the five seeds (Table 2 and Figure 7), applying the pilot-validated $CE_{seed} > 0$ gate via **VLEG-CG-Mal** reduces the overall FPR by −9.35 pp (from 48.78% to 39.43% ± 4.68%) and raises Macro-F1 by +3.64 pp (to 63.32% ± 2.10%) while retaining a higher attack recall (72.11% ± 8.20%) than Raw retrieval (71.39%). Applying symmetric gating across both classes (**VLEG-CG**) reduces the overall FPR by −10.92 pp (to 37.86% ± 7.16%) and recovers Macro-F1 to 63.55% ± 2.49%.

Because counterexample gating falls back to raw admitted flows when retrieved rules have $CE_{seed} > 0$, a crucial question is how often verbal rules are retained and whether retained rules improve or degrade classification on the un-gated queries. Table 4 disentangles performance on the **Un-Gated Subset** (where verbal rules are displayed) versus the **Gated Subset** (where over-generalized rules are stripped) across three gating granularities:

1. **All-Rule Gate (VLEG-CG,** 12.3% **rules retained):** Because 59.2% of individual rules have $CE_{seed} > 0$, requiring all $k = 3$ retrieved rules to have zero within-seed counterexamples retains verbal rules on 122.8/1000 (12.3%) of test queries. Crucially, on this 12.3% un-gated subset, **VLEG with zero-counterexample rules outperforms Raw retrieval on the exact same queries by +1.57 pp in Macro-F1 (64.60% versus 63.03%) while lowering the FPR by −3.00 pp (33.11% versus 36.11%)**. On the remaining 87.7% (877.2/1000) gated queries, fallback reduces FPR from 50.71% to 38.01% (−12.70 pp).

Table 4. Subset disentanglement of public-flow performance across the five formal seeds (1000 test queries/seed), separating the **Un-Gated Subset** (queries where verified verbal rules are retained in the prompt) from the **Gated Subset** (queries where over-generalized rules are stripped and inference falls back to raw admitted flows). Within each subset, VLEG (with rules) is compared against Raw admitted flows (without rules) on the exact same test queries.

		Un-Gated Subset (Rules Retained)			Gated Subset (Over-Generalized Rules)		
Gating Policy	**Rule-Retained Queries**	**Method**	**Macro-F1**	**FPR**	**Method**	**Macro-F1**	**FPR**
VLEG-CG (All-rule CE_{seed}=0)	**122.8 / 1000 (12.3%)**	**VLEG (rules kept)**	**64.60%**	**33.11%**	VLEG (unscreened)	58.40%	50.71%
		Raw (no rules)	63.03%	36.11%	**VLEG-CG (fallback)**	**62.87%**	**38.01%**
VLEG-CG-Mal (Malicious CE_{seed}=0)	**409.8 / 1000 (41.0%)**	**VLEG (rules kept)**	65.87%	26.16%	VLEG (unscreened)	50.35%	67.87%
		Raw (no rules)	**66.27%**	**24.92%**	**VLEG-CG-Mal (fallback)**	**59.07%**	**48.91%**
VLEG-CG-ActiveMal (Active firing CE gate)	**728.2 / 1000 (72.8%)**	VLEG (rules kept)	61.81%	38.22%	VLEG (unscreened)	43.09%	83.92%
		Raw (no rules)	**63.91%**	**36.34%**	**ActiveMal (fallback)**	**58.45%**	**49.98%**

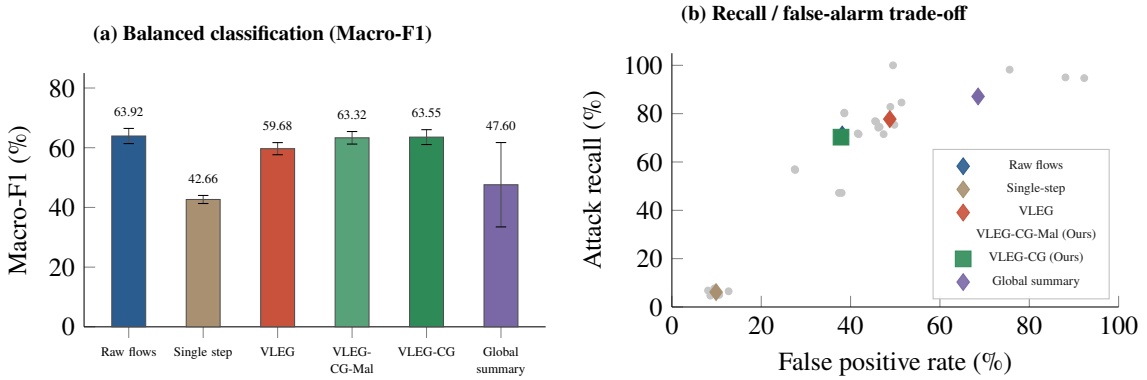


Figure 7. Public-flow results across five historical-memory seeds on UNSW-NB15. (a) Mean Macro-F1 with sample-standard-deviation bars; (b) attack recall versus false positive rate (faint points denote individual seeds; large markers denote 5-seed means).

2. **Malicious-Rule Gate (VLEG-CG-Mal,** 41.0% **rules retained):** Gating only on over-generalized malicious rules retains verified verbal rules on 409.8/1000 (41.0%) of queries, where VLEG achieves 65.87% Macro-F1 and a low 26.16% FPR (within 0.40 pp of Raw retrieval's 66.27% Macro-F1 while supplying explicit verbal rules). Conversely, on the 59.0% (590.2/1000) gated queries where an over-generalized malicious rule was retrieved, unscreened VLEG suffers a severe **67.87% FPR and 50.35% Macro-F1**, which VLEG-CG-Mal improves by −18.96 pp in FPR (to 48.91%) and +8.72 pp in Macro-F1 (to 59.07%).

3. **Active-Trigger Malicious Gate (VLEG-CG-ActiveMal,** 72.8% **rules retained):** If gating is triggered only when a retrieved malicious rule with $CE_{seed} > 0$ *actively fires* ($\pi(r_i^*)(x_{test}) = 1$) on the current test flow, verbal rules are retained on 728.2/1000 (72.8%) of queries (62.15% overall Macro-F1, 40.33% overall FPR). On the 27.2% (271.8/1000) of queries where an over-generalized malicious rule actively fires on the test flow, unscreened VLEG exhibits an extreme **83.92% FPR (43.09% Macro-F1)**, which gating reduces by **−33.94 pp** (to 49.98% FPR and 58.45% Macro-F1).

Table 4 thus provides direct mechanistic evidence: local reflections that satisfy $CE_{seed} = 0$ across both classes genuinely improve classification over raw flows (64.60% vs. 63.03% Macro-F1), whereas standard VLEG's aggregate deficit is driven specifically by the subset of queries where $CE_{seed} > 0$ rules fire.

Table 5 records the consolidation and query costs. Across the 5000 predictions per method, there were 29 unparseable outputs for unverified single-step reflection and 30 for standard VLEG (counted as errors). VLEG consolidation averages 120.4 LLM calls and 40,094 tokens (419.43 s) per 32-flow seed, whereas deterministic counterexample compilation adds less than 10 ms of CPU time and zero LLM tokens.

4.8. Comparison with Symbolic Decision Trees (CART), Ensembles (RF, GBDT), and LLM-IDS

On the 32-flow few-shot splits, shallow CART decision trees and tree ensembles achieve higher Macro-F1 (65.84%–70.25%) **than in-context LLM methods** (63.55%–63.92%) **while exhibiting comparable false positive rates** (36.53%–40.66% **versus** 37.86%–39.43% **for VLEG-CG); however, CART splits are restricted to**

Table 5. Observed public-flow memory construction and query costs (means across five memory seeds, 32 historical flows per seed). VLEG-CG and VLEG-CG-Mal perform deterministic Python predicate screening in < 0.01 s with zero additional LLM calls or tokens.

Condition	Construction Calls	Construction Tokens	Construction (s)	Query Mean (s)	Query p95 (s)
Raw flow retrieval	0	0	0	2.12	2.67
Unverified single-step	32.0	9405	126.54	2.23	2.75
VLEG / VLEG-CG (K=3)	120.4	40,094	419.43	1.74	2.68
Raw on admitted flows	120.4	40,094	419.43	0.95	1.19
Global summary	1.0	5947	23.21	1.21	1.30

global axis-aligned thresholds and cannot express relational feature predicates or semantic triage rationales. To compare VLEG-CG against both interpretable symbolic rule learners and ensemble classifiers, Table 2 reports shallow CART Decision Trees (`max_depth=3`, averaging 4.4 leaves, and unconstrained CART, averaging 5.2 leaves), Random Forest (200 trees), and GBDT (100 trees) across the exact same five 32-flow seeds and the full 88,699-flow cleaned training set:

1. **Shallow Decision Trees vs. VLEG-CG in the 32-Flow Regime:** Trained on the 32 support flows per seed, a depth-3 CART tree achieves 69.39% ± 1.76% Macro-F1 (83.38% recall, 36.53% ± 6.76% FPR) and an unconstrained CART tree achieves 70.25% ± 0.64% Macro-F1 (86.41% recall, 37.13% ± 5.08% FPR), outperforming VLEG-CG (63.55% Macro-F1, 37.86% FPR) by 5.84–6.70 pp in Macro-F1 at negligible CPU cost. This gap arises because CART optimizes split thresholds *globally* across all 32 training flows simultaneously, guaranteeing zero within-seed counterexamples ($CE_{seed} = 0$) on pure leaves by construction, whereas VLEG induces rules *locally* from individual episodes (e_i) and uses $CE_{seed} = 0$ only as a post-hoc filter.

2. **Structural Expressiveness and Interpretability Trade-offs (CART vs. LLM Rules):** Examining the induced rules on Seed `20261020` highlights a qualitative difference between classical tree induction and LLM reflection. The depth-3 CART tree partitions the entire 32-flow space using three univariate axis-aligned numeric splits (`sttl <= 46.50` → `BENIGN`; `sttl > 46.50 AND dinpkt <= 67.08` → `MALICIOUS`; `dinpkt > 67.08 AND djit > 191.89` → `MALICIOUS`). While statistically effective on the sample, these splits provide no protocol semantics and cannot compare two features directly. By contrast, VLEG's verified reflection on flow `622f0c59...` states: `if proto == "tcp" and state == "FIN" and dur < 1 and sbytes > dbytes, then label = MALICIOUS, due to asymmetric byte transfer in a short-lived TCP connection`. VLEG naturally expresses cross-feature relational predicates (`sbytes > dbytes`) paired with domain-grounded protocol explanations and supports modular per-episode insertion or deletion without refitting a global tree.

3. **Impact of 19-Feature Deduplication on Full-Train Models (88,699 Flows):** When trained on all 88,699 deduplicated training fingerprints across five random seeds, full-train Random Forest achieves 82.06%±0.36% accuracy (81.31% ± 0.36% Macro-F1, 91.99% recall, **22.99% ± 0.55% FPR**), full-train GBDT achieves 76.50% accuracy (76.25% Macro-F1, 98.22% recall, 34.54% FPR), and a depth-5 CART tree achieves 77.20% accuracy (76.81% Macro-F1, 31.98% FPR). Thus, once the 79,746 duplicate training bursts and 33,161 duplicate test rows are removed, even full-train models face a 23%–35% FPR on unique boundary fingerprints, and all 32-flow few-shot baselines (CART, RF, GBDT, Raw retrieval, and VLEG-CG) converge to a comparable FPR range of 36.53%–40.66%. Compared with recent RAG-based LLM-IDS frameworks [5,6], VLEG-CG demonstrates that local verbal rules must be paired with counterexample screening to remain competitive with raw feature retrieval.

4.9. Limitations

The symbolic evaluations use a small 16-scene population, while the public-flow study evaluates a deduplicated UNSW-NB15 testbed partition. The latter employs standardized feature-indexed retrieval across local memory conditions to ensure a controlled, apples-to-apples mechanism comparison. Symbolic Levels 1 and 2 evaluate seen combinations, and Level 3 combines seen and withheld combinations; their accuracy must not be interpreted as held-out network generalization. The public-flow experiment is bounded by 19-feature deduplication and

conflict exclusion, missing raw packet payloads, a single 70B quantized model (`Q4_K_M`) and test partition, fixed method execution order, and automated compiler screening rather than human SOC analyst trials.

Furthermore, the synthetic rule is expressible as a single background-conditioned statement, so it tests conditional rule induction rather than an impossibility of global abstraction. The 7B and 70B symbolic comparisons also change model family (Qwen2.5 versus Llama-3.3) and therefore cannot isolate parameter scale causally. Observed rank reversals and equal diagnostic rates are descriptive findings in these samples rather than universal equivalence claims.

Finally, while our deterministic counterexample compiler Ψ screens admitted rules against the remaining 31 historical episodes within each seed ($\mathcal{E}_{\text{seed}} \setminus \{e_i\}$), a 32-episode support bank is a small sample: achieving zero within-seed counterexamples ($\text{CE}_{\text{seed}} = 0$) reduces false alarms by 9.35–10.92 pp, but 59.2% of locally generated rules are gated out, leaving VLEG-CG 4.87–6.70 pp behind 32-flow CART and Random Forest in overall Macro-F1. Feeding counterexamples back into the generator during Phase 1 consolidation to actively rewrite over-generalized rules rather than falling back to raw flows is therefore a key next step.

5. Conclusions

This paper presented Verified Localized Episodic Grounding (VLEG) and its counterexample-gated extension (VLEG-CG), an auditable memory-consolidation and hybrid-inference framework for few-shot LLM-assisted network traffic analysis. Across a 16-scene symbolic sanity check and a 25,000-prediction benchmark on 1000 deduplicated UNSW-NB15 test fingerprints, our study yields four principal conclusions:

1. **Multi-Step Source Verification is Required to Prevent Label Inversion:** Unverified single-step reflection mislabels 95.0% (76/80) of malicious historical flows as benign, causing test attack recall to drop to 6.17% (42.66% Macro-F1). Crucially, single-pass label feedback at $j = 1$ fixes only 14.5% (11/76) of these rejected malicious flows, whereas a second verification-enforced revision at $j = 2$ recovers 63/76 (82.9%), restoring 100% source-label agreement on admitted episodes (158/158) and lifting attack recall to 77.74% (59.68% Macro-F1).

2. **Single-Episode Fidelity Does Not Prevent Cross-Flow Over-Generalization:** Despite 100% source-label agreement and $40,094$ construction tokens per seed, unscreened VLEG trails zero-cost raw flow retrieval (63.92% Macro-F1) by 4.24 pp because 59.2% (93/157) of compiled rules match contrary-label flows within their own 32-flow seed (72.0%, 113/157 across all 160 pooled flows), inflating the false positive rate to 48.78%.

3. **Within-Seed Counterexample Gating Restores Precision and Outperforms Raw Retrieval on Un-Gated Queries:** Validated on a disjoint 100-query pilot set, screening admitted rules with our deterministic counterexample compiler (VLEG-CG and VLEG-CG-Mal) at zero extra LLM cost reduces the test FPR by 9.35–10.92 pp (to 37.86%–39.43%) and recovers overall Macro-F1 to 63.32%–63.55%. On the un-gated query subset where all retrieved rules satisfy $\text{CE}_{\text{seed}} = 0$ (12.3% of queries under VLEG-CG), retaining verified rules outperforms raw retrieval on the exact same queries (64.60% versus 63.03% Macro-F1, −3.00 pp FPR), while fallback on the gated subset prevents severe false-alarm spikes (67.87%–83.92% FPR).

4. **Positioning Relative to Symbolic Decision Trees and Tabular Ensembles:** On the same 32-flow splits, shallow CART decision trees (69.39%–70.25% Macro-F1, 36.53%–37.13% FPR) and Random Forests (68.42% Macro-F1, 40.66% FPR) achieve higher discriminative Macro-F1 than in-context LLM methods (63.55%–63.92%) because they optimize feature thresholds globally across all 32 episodes, whereas full-train Random Forests (82.06% accuracy, 22.99% FPR on 88,699 unique flows) provide the strongest line-rate filter. Conversely, VLEG-CG provides relational feature predicates (e.g., `sbytes > dbytes`) and protocol-level verbal explanations for human triage that axis-aligned tree splits cannot express. Future work will inject within-seed counterexample traces directly into the Phase 1 generator prompt so that candidate rules are iteratively specialized to achieve $\text{CE}_{\text{seed}} = 0$ prior to memory commitment.

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References

[1] Brown, T.B.; Mann, B.; Ryder, N.; Subbiah, M.; Kaplan, J.; Dhariwal, P.; Neelakantan, A.; Shyam, P.; Sastry, G.; Askell, A.; et al. Language Models are Few-Shot Learners. *Adv. Neural Inf. Process. Syst.* **2020**, *33*, 1877–1901.

[2] Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; Yih, W.T.; Rocktäschel, T.; et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Adv. Neural Inf. Process. Syst.* **2020**, *33*, 9459–9474.

[3] Lin, X.; Xiong, G.; Gou, G.; Li, Z.; Shi, J.; Yu, J. ET-BERT: A Contextualized Datagram Representation with Pre-training Transformers for Encrypted Traffic Classification. In Proceedings of the ACM Web Conference 2022, Virtual Event, 25–29 April 2022; pp. 633–642.

[4] Sharafaldin, I.; Lashkari, A.H.; Ghorbani, A.A. Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization. In Proceedings of the International Conference on Information Systems Security and Privacy (ICISSP), Funchal, Portugal, 22–24 January 2018; pp. 108–116.

[5] Houssel, P.R.B.; Singh, P.; Layeghy, S.; Portmann, M. Towards Explainable Network Intrusion Detection Using Large Language Models. *arXiv* **2024**, arXiv:2408.04342.

[6] Islam, M.S.; Jaimes, L.G.; Dina, A.S. MA-IDS: Multi-Agent RAG Framework for IoT Network Intrusion Detection with an Experience Library. *arXiv* **2026**, arXiv:2604.05458.

[7] An, C.; Zhang, J.; Zhong, M.; Li, L.; Gong, S.; Luo, Y.; Xu, J.; Kong, L. Why Does the Effective Context Length of LLMs Fall Short? *arXiv* **2024**, arXiv:2410.18745.

[8] Liu, N.F.; Lin, K.; Hewitt, J.; Paranjape, A.; Bevilacqua, M.; Petroni, F.; Liang, P. Lost in the Middle: How Language Models Use Long Contexts. *Trans. Assoc. Comput. Linguist.* **2024**, *12*, 157–173.

[9] Packer, C.; Wooders, S.; Lin, K.; Fang, V.; Patil, S.G.; Stoica, I.; Gonzalez, J.E. MemGPT: Towards LLMs as Operating Systems. *arXiv* **2023**, arXiv:2310.08560.

[10] Wang, W.; Dong, L.; Cheng, H.; Liu, X.; Yan, X.; Gao, J.; Wei, F. Augmenting Language Models with Long-Term Memory. *Adv. Neural Inf. Process. Syst.* **2023**, *36*, 74530–74543.

[11] Wang, Y.; Gao, Y.; Chen, X.; Jiang, H.; Li, S.; Yang, J.; Yin, Q.; Li, Z.; Li, X.; Yin, B.; et al. MEMORYLLM: Towards Self-Updatable Large Language Models. In Proceedings of the 41st International Conference on Machine Learning (ICML), Vienna, Austria, 21–27 July 2024; Volume 235, pp. 50453–50466.

[12] Jimenez Gutierrez, B.; Shu, Y.; Gu, Y.; Yasunaga, M.; Su, Y. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. *Adv. Neural Inf. Process. Syst.* **2024**, *37*, 59532–59569.

[13] Maharana, A.; Lee, D.H.; Tulyakov, S.; Bansal, M.; Barbieri, F.; Fang, Y. Evaluating Very Long-Term Conversational Memory of LLM Agents. *arXiv* **2024**, arXiv:2402.17753.

[14] Wu, Y.; Liang, S.; Zhang, C.; Wang, Y.; Zhang, Y.; Guo, H.; Tang, R.; Liu, Y. From Human Memory to AI Memory: A Survey on Memory Mechanisms in the Era of LLMs. *arXiv* **2025**, arXiv:2504.15965.

[15] Zhang, Z.; Bo, X.; Ma, C.; Li, R.; Chen, X.; Dai, Q.; Zhu, J.; Dong, Z.; Wen, J.R. A Survey on the Memory Mechanism of Large Language Model Based Agents. *arXiv* **2024**, arXiv:2404.13501.

[16] Shinn, N.; Cassano, F.; Gopinath, A.; Narasimhan, K.R.; Yao, S. Reflexion: Language Agents with Verbal Reinforcement Learning. *Adv. Neural Inf. Process. Syst.* **2023**, *36*, 8634–8652.

[17] Wang, G.; Xie, Y.; Jiang, Y.; Mandlekar, A.; Xiao, C.; Zhu, Y.; Fan, L.; Anandkumar, A. Voyager: An Open-Ended Embodied Agent with Large Language Models. *Trans. Mach. Learn. Res.* **2024**, *2024*, 1–32.

[18] Agarwal, A.K.; Yamada, M. Hypothesis-Driven Reasoning for Large Language Models. In Proceedings of the Fortieth AAAI Conference on Artificial Intelligence (AAAI-26), Philadelphia, PA, USA, 24 February–3 March 2026; pp. 1686–1693.

[19] He, K.; Zhang, M.; Yan, S.; Wu, P.; Chen, Z. IDEA: Enhancing the Rule Learning Ability of Large Language Model Agent through Induction, Deduction, and Abduction. In Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL), Bangkok, Thailand, 11–16 August 2024.

[20] Zhang, Y.; Feng, S.; Tan, C. Active Example Selection for In-Context Learning. *arXiv* **2022**, arXiv:2211.04486.

[21] Purohit, K.; V, V.; Bhattacharya, S.; Anand, A. Sample Efficient Demonstration Selection for In-Context Learning. In Proceedings of the 42nd International Conference on Machine Learning (ICML), Vancouver, BC, Canada, 13–19 July 2025.

[22] Zhong, W.; Guo, L.; Gao, Q.; Ye, H.; Wang, Y. MemoryBank: Enhancing Large Language Models with Long-Term Memory. *Proc. AAAI Conf. Artif. Intell.* **2024**, *38*, 19724–19731.

[23] Park, J.S.; O'Brien, J.C.; Cai, C.J.; Morris, M.R.; Liang, P.; Bernstein, M.S. Generative Agents: Interactive Simulacra of Human Behavior. In Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST), San Francisco, CA, USA, 29 October–1 November 2023; pp. 1–22.

[24] Xu, W.; Liang, Z.; Mei, K.; Gao, H.; Tan, J.; Zhang, Y. A-MEM: Agentic Memory for LLM Agents. *arXiv* **2025**, arXiv:2502.12110.

[25] Kim, S.H.; Ong, K.T.; Kwon, T.; Kim, N.; Ka, K.; Bae, S.; Jo, Y.; Hwang, S.; Lee, D.; Yeo, J. Towards Lifelong Dialogue Agents via Timeline-Based Memory Management. In Proceedings of the North American Chapter of the Association for Computational Linguistics (NAACL), Mexico City, Mexico, 16–21 June 2024.

[26] Kim, S.; Phunyaphibarn, P.; Ahn, D.; Kim, S. Playgrounds for Abstraction and Reasoning. In Proceedings of the NeurIPS Workshop on Neuro Causal and Symbolic AI (nCSI), New Orleans, LA, USA, 2 December 2022.

[27] Lee, K.I.; Koh, H.; Lee, D.; Yoon, S.; Kim, M.; Jung, K. Generating Diverse Hypotheses for Inductive Reasoning. *arXiv* **2024**, arXiv:2412.13422.

[28] Qiu, L.; Jiang, L.; Lu, X.; Sclar, M.; Pyatkin, V.; Bhagavatula, C.; Wang, B.; Kim, Y.; Choi, Y.; Dziri, N.; et al. Phenomenal Yet Puzzling: Testing Inductive Reasoning Capabilities of Language Models with Hypothesis Refinement. In Proceedings of the International Conference on Representation Learning (ICLR), Vienna, Austria, 7–11 May 2024; pp. 1452–1481.

[29] Zhou, Y.; Liu, H.; Srivastava, T.; Mei, H.; Tan, C. Hypothesis Generation with Large Language Models. In Proceedings of the EMNLP Workshop on NLP for Science, Miami, FL, USA, 15 November 2024.

[30] Kim, T.; Cochez, M.; François-Lavet, V.; Neerincx, M.; Vossen, P. A Machine with Short-Term, Episodic, and Semantic Memory Systems. *Proc. AAAI Conf. Artif. Intell.* **2023**, *37*, 48–56.

[31] Li, J.; Li, J. Memory, Consciousness and Large Language Model. *arXiv* **2024**, arXiv:2401.02509.

[32] Wang, Y.; Chen, X. MIRIX: Multi-Agent Memory System for LLM-Based Agents. *arXiv* **2025**, arXiv:2507.07957.

[33] Manakul, P.; Liusie, A.; Gales, M. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), Singapore, 6–10 December 2023; pp. 9004–9017.

[34] Dhuliawala, S.; Komeili, M.; Xu, J.; Raileanu, R.; Li, X.; Celikyilmaz, A.; Weston, J. Chain-of-Verification Reduces Hallucination in Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*; Association for Computational Linguistics: Bangkok, Thailand, 2024; pp. 3563–3578.

[35] Madaan, A.; Tandon, N.; Gupta, P.; Hallinan, S.; Gao, L.; Wiegreffe, S.; Alon, U.; Dziri, N.; Prabhumoye, S.; Yang, Y.; et al. Self-Refine: Iterative Refinement with Self-Feedback. *Adv. Neural Inf. Process. Syst.* **2023**, *36*, 46534–46594.

[36] Gou, Z.; Shao, Z.; Gong, Y.; Shen, Y.; Yang, Y.; Duan, N.; Chen, W. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. In Proceedings of the International Conference on Learning Representations (ICLR), Vienna, Austria, 7–11 May 2024.

[37] Moustafa, N.; Slay, J. UNSW-NB15: A Comprehensive Data Set for Network Intrusion Detection Systems (UNSW-NB15 Network Data Set). In Proceedings of the Military Communications and Information Systems Conference (MilCIS), Canberra, Australia, 10–12 November 2015; pp. 1–6.